5. Find the tangent and normal to the following curves at the given points on the curve.

(i) $y = x^2 - x^4$ at $(1,0)$ (ii) $y = x^4 + 2e^x$ at $(0,2)$

(iii) $y = x\sin x$ at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ (iv) $x = \cos t, y = 2\sin^2 t$ at $t = \frac{\pi}{3}$

6. Find the equations of the tangents to the curve $y = 1 + x^3$ for which the tangent is orthogonal with the line $x + 12y = 12$.

7. Find the equations of the tangents to the curve $y = \frac{x+1}{x-1}$ which are parallel to the line $x + 2y = 6$.

8. Find the equation of tangent and normal to the curve given by $x = 7\cos t$ and $y = 2\sin t, t \in \mathbb{R}$ at any point on the curve.

9. Find the angle between the rectangular hyperbola $xy = 2$ and the parabola $x^2 + 4y = 0$.

10. Show that the two curves $x^2 - y^2 = r^2$ and $xy = c^2$ where $c, r$ are constants, cut orthogonally.

## 7.3 Mean Value Theorem

Mean value theorem establishes the existence of a point, in between two points, at which the tangent to the curve is parallel to the secant joining those two points of the curve. We start this section with the statement of the intermediate value theorem as follows :

**Theorem 7.1 (Intermediate value theorem)**

If $f$ is continuous on a closed interval $[a,b]$, and $c$ is any number between $f(a)$ and $f(b)$ inclusive, then there is at least one number $x$ in the closed interval $[a,b]$, such that $f(x) = c$.

### 7.3.1 Rolle's Theorem

**Theorem 7.2 (Rolle's Theorem)**

Let $f(x)$ be continuous on a closed interval $[a,b]$ and differentiable on the open interval $(a,b)$ If $f(a) = f(b)$, then there is at least one point $c \in (a,b)$ where $f'(c) = 0$.

Geometrically this means that if the tangent is moving along the curve starting at $x = a$ towards as in Fig 7.2 $x = b$ then there exists a $c \in (a,b)$ at which the tangent is parallel to the $x$-axis.

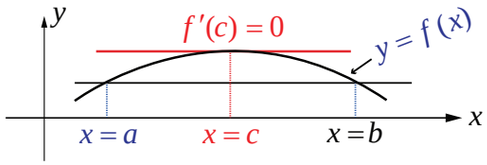


Fig.7.12

**Example 7.19**

Compute the value of '$c$' satisfied by the Rolle's theorem for the function

$f(x) = x^2(1-x)^2, x \in [0,1]$.

**Solution**

Observe that, $f(0) = 0 = f(1), f(x)$ is continuous in the interval $[0,1]$ and is differentiable in $(0,1)$. Now,

$$f'(x) = 2x(1-x)(1-2x).$$



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